

Design of a Physical-Only Proprioceptive Recognition System for Twenty Household Objects with a LEAP Hand

G. Sai Pranav, Harish J. Palanthandalam-Madapusi

Abstract

Robotic hands commonly use vision or dedicated tactile sensors to recognize objects. This study examines whether a physical LEAP Hand can identify household items using only the joint angles reached after controlled grasps. Motor-current thresholds stop an entire finger when resistance is detected. Final angles from an ordered sequence of grips form the input to a compact neural classifier; all training measurements come from the physical hand. The system is designed for 20 household-object classes, with three sequential grips providing 48 final joint-angle measurements per trial. A 24-unit hidden layer maps these measurements to class probabilities. Physical trials are augmented only after real measurements have been separated into training and validation folds. A distance and confidence rule permits uncertain or unknown outputs, and repeated trials can be combined when one reading is inconclusive. Evaluation requires held-out trials across objects and sessions, as well as objects excluded from training to test unknown-object rejection.

Keywords: proprioception, active perception, LEAP Hand, household object recognition, neural network, physical grasping

1. Introduction

A robot can learn about an object by interacting with it. For a dexterous hand, the joint positions reached while closing around an object depend on the surfaces that interrupt finger motion. These positions may therefore contain information about the object even without an external camera or dedicated tactile skin. This use of the hand’s internal state is known as proprioceptive perception. The LEAP Hand is a dexterous, anthropomorphic hand developed for robot learning and manipulation (Shaw et al., 2023).

Previous work has shown that joint information can support object recognition. Vásquez and Perdereau (2017) developed proprioceptive shape signatures from robotic-hand joint angles and studied shape identification across object sizes and poses. Sintov and Meir (2023) used grasp information for kinesthetic object recognition without dedicated tactile sensors. More recently, Kutsuzawa et al. (2024) investigated learning-based shape and stiffness estimation from joint angles and velocities in a multi-fingered hand. These studies motivate a physical-hand experiment using a simple observation: the final joint angles of a repeatable grip sequence.

The main research question is whether final angles from the physical LEAP Hand can distinguish 20 enrolled household objects. A second question is whether combining repeated grip sequences improves decisions that are uncertain after one sequence. The model predicts object identity, rather than object size. Unless multiple examples of a household category are collected, each label refers to one physical item; recognition of another item of the same kind is a separate generalisation question.

The implementation provides repeatable grips, a resumable physical-data collection procedure, a neural classifier, a nearest-trial comparison method, and an interface for running the hand and viewing the results. The study is organized around recognition among 20 enrolled household objects.

2. Experimental Setup and Methodology

2.1. *Physical hand and controlled grips*

The protocol uses a four-fingered LEAP Hand with 16 actuated joints (Shaw et al., 2023). The joint vector is ordered as index joints 0–3, middle joints 4–7, ring joints 8–11, and thumb joints 12–15. All angles are expressed in degrees. Each object is placed near a consistent point relative to the palm, with a prescribed variation of 2 cm in position and 10° in orientation. Every object receives the same programmed grip sequence. Placement is a controlled experimental condition rather than a property inferred by the model.

For each configured grip, the hand moves to that grip’s starting posture. Its 16 joints then advance toward specified targets using the per-joint step sizes in the experiment settings. During closure, motor current served as an indicator of resistance. If a moving joint’s signed current exceeds its configured threshold, all four joints of that finger stop at their measured positions. The remaining fingers continue until resistance or an angle target

stops them. A grip ends when all four fingers have stopped and is aborted after at most 250 iterations. The hand moves to the next grip’s starting posture before continuing the sequence.

After each grip, the program reads its final 16 joint angles, retrying for up to 10 seconds if position reporting is temporarily unavailable. The final vectors are concatenated in grip order. The current settings contain three grips, yielding 48 input angles per physical trial.

During collection, the final angles and time-series log are saved before the hand returns to its starting posture. Motor currents control the movement but are not classifier inputs. No camera images or tactile arrays are used as recognition features.

2.2. Twenty-object physical dataset

Each training record consists of a household-object label and one concatenated final-angle vector,

$$\mathbf{q}_i = [\mathbf{q}_i^{(1)}, \dots, \mathbf{q}_i^{(G)}] \in \mathbb{R}^{16G},$$

where G is the number of configured grips and each $\mathbf{q}_i^{(g)}$ contains the 16 final joint angles from grip g . The data-collection program stores every completed trial with its final angles and time-series log. It also saves a collection plan before recording begins, so interrupted sessions can resume without discarding earlier trials. Additional trials and new classes can be added later. A mistaken trial may be retaken; the previous record is archived after a complete replacement has been checked.

The study design specifies 20 distinct household objects, with five to ten independent physical trials per object, or 100–200 labeled trials. The selected objects should be graspable within the hand’s configured motion limits and should vary in geometry and size. Their names, dimensions, and placement orientation are recorded with the dataset. Repeated trials include reseating the object between measurements. A separate later session is reserved to test changes in placement and hand state.

2.3. Neural network and training

The neural network receives only the $16G$ final joint angles. Each angle is normalized by the start and target angles of its grip and joint:

$$x_{g,j} = \frac{q_{g,j} - a_{g,j}}{b_{g,j} - a_{g,j}},$$

where $a_{g,j}$ and $b_{g,j}$ are the start and target angles for joint j in grip g . A joint with equal start and target values is assigned a normalized value of zero.

The classifier has $16G$ input units, one fully connected 24-unit hidden layer with a hyperbolic tangent activation, and one output for each enrolled object. A softmax operation converts the outputs to class probabilities. For C classes, the model has $384G + 24 + 25C$ trainable parameters. With three grips and 20 classes, it has 1,676 parameters. The model does not predict object size.

Training uses full-batch Adam optimization with a learning rate of 0.01, weight decay of 0.001, and a limit of 500 epochs. Each real trial is retained and supplemented with 20 noisy copies during training. Independent zero-mean Gaussian noise with a standard deviation of 0.5° is added to each angle. A second 0.5° Gaussian perturbation is shared by the four joints of each finger within each grip. These artificial copies approximate small measurement and grasp variation; they are not counted as additional physical trials. Samples are weighted inversely to the number of real trials in their class to reduce the effect of unequal class sizes.

2.4. Validation and recognition decisions

The implemented evaluation uses up to three folds, with real trials from every class assigned to different folds before noise is added. Noise augmentation is performed only on each training fold. The validation fold therefore contains measurements from the physical hand, not perturbed copies of training samples. The best epoch in each fold is selected by validation cross-entropy, with training stopped after 80 epochs without improvement. The final model is then trained on all real trials for the median number of selected epochs.

The class probabilities are adjusted with a temperature selected from the held-out folds. We also compute the distance from a new normalized angle vector to the nearest saved physical training grasp. A prediction is reported as *unknown* when this distance exceeds 1.5 times the 95th percentile of the validation distances. Otherwise, it is reported as *uncertain* when the largest class probability is below 0.75 or the difference between the two largest probabilities is below 0.20. A known class is returned only when these checks pass. These thresholds are calibrated from known-class validation trials. Their ability to reject genuinely unseen objects requires a separate evaluation.

If a neural prediction remains uncertain, the program can request another trial of the same object, up to three trials. Each trial runs the configured grip sequence. The class probabilities are averaged across trials, while the nearest-reference distances are combined using their median. The resulting decision is saved with each trial. A separate nearest-trial method assigns the class of the saved physical grasp with the smallest Euclidean distance in degrees. That method supplies a comparison baseline; it does not produce probabilities or an unknown-object decision.

2.5. *Software implementation*

The grip and physical-only training programs are part of Ex-ARM (Pranav, 2026). The terminal interface can collect, resume, extend, and retake physical trials, train either model, and run the hand for recognition. The presentation window shows the final decision, the neural probabilities for every enrolled class, and a visualization driven by the measured hand pose. The classifier itself uses only recorded physical angles.

3. Evaluation Protocol and Discussion

3.1. *Twenty-object evaluation*

The principal outcome is raw classification accuracy across the 20 enrolled household objects. Each held-out physical trial contributes one true class and one predicted class. A 20-by-20 confusion matrix and per-class recall identify object pairs that produce similar final joint postures. Macro-averaged recall is included alongside overall accuracy so that classes with fewer trials do not disappear inside an aggregate score.

The decision rule introduces two additional outcomes: *uncertain* and *unknown*. Evaluation therefore reports coverage, defined as the fraction of trials assigned a known class, and accuracy among accepted predictions. These measures are reported together; a high accepted accuracy by itself could result from rejecting many difficult trials. To assess unknown-object rejection, some physical objects must be excluded entirely from model training and treated as novel at test time.

A later collection session should be held out until the network, normalization, and decision thresholds are fixed. This test measures how object placement, motor calibration, and hand state affect recognition beyond the within-session validation folds. The same held-out objects can then be classified after one grip sequence and after repeated sequences. This paired

comparison measures whether another physical observation improves the decision.

3.2. Interpretation and limitations

The final joint configuration contains information about where the fingers were stopped, but it is not a unique description of an object’s shape. Items of different appearance may produce similar joint angles when held at the same palm location. The use of several grip patterns may expose more of the object, while also introducing additional variation if the object shifts between grips. These effects must be evaluated on recorded physical trials, rather than inferred from the network architecture.

Gaussian angle perturbations can help tolerate small measurement changes but do not create new independent physical grasps. They cannot reproduce every change in placement, friction, compliance, or motor-current response. The study therefore relies on repeated real trials and later-session testing. If each class is represented by only one physical exemplar, recognition of another item from the same category remains a separate question.

4. Conclusion

This work specifies a physical-only proprioceptive recognition system for 20 household objects using a LEAP Hand. Current thresholds stop contacted fingers, and the final angles from three sequential grips form a 48-value input to a compact neural classifier. The software supports resumable collection, trial retakes, class probabilities, uncertain or unknown decisions, and repeated recognition attempts.

The evaluation protocol includes held-out physical accuracy, per-class performance, decision coverage, and tests on later sessions and unseen objects. These measurements are required before a quantitative claim about 20-object recognition can be made.

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